The Alaoglu–Erdős Conjecture

Rotation Discriminants, Matched-Gap Energy, and Radical Norm Compression

An independent continuation of the supplied unified report

Scope. This document does not repeat the supplied report's twenty-seven continuations, its Lean inventory, or its proofs of the primitive-generator theorem. It imports only the exact conditional free-monoid structure and the finite-type consequence of Baker's theorem, then develops a different object: the Vandermonde product of the compact rotation orbit. The resulting formulas expose an exact second-order imbalance between the base-2 and base-3 integer gap products and a concrete cross-prime transport problem.

Trust convention. Every imported statement is marked **[imported]**. Every theorem proved here is marked **[new proof]**. Numerical values are marked **[computation]**. No Lean verification of the new material is claimed; a formalization blueprint is supplied. The full Alaoglu–Erdős conjecture is not claimed proved.

August 22, 2026

Abstract

Let $\vartheta = \log_2 3$. The Alaoglu–Erdős conjecture says that $2^x, 3^x \in \mathbb{Z}$ implies $x \in \mathbb{Z}$. Conditional on failure, the supplied report proves that there is a least nonintegral solution β and integers $M = 2^\beta$, $A = 3^\beta$ such that every solution is $n + k\beta$. Writing $r_k = \lfloor k\beta \rfloor$ and $t_k = \{k\beta\}$, the compact orbit is $(2^{t_k}, 3^{t_k}) = (M^k/2^{r_k}, A^k/3^{r_k})$.

This continuation studies the products of all pairwise differences of that orbit. It proves an exact lag-by-lag factorization into two Sudler-type factors, an exact structural-prime denominator formula, an integer orbit-polynomial discriminant identity, and a logarithmic energy limit with a closed form involving $\zeta(3)$ and Li_3. The two bases can then be compared pointwise. A strictly decreasing universal function

$$\Phi(u) = \log_3(3^u - 1) - \log_2(2^u - 1)$$

satisfies $\Phi(u) > \log_3 2$ on $(0, 1)$. Consequently every synchronized near-power gap obeys

$$|A^d - 3^h| > 2\,|M^d - 2^h|^\vartheta \qquad (h \in \{\lfloor d\beta \rfloor, \lceil d\beta \rceil\}),$$

and the complete gap products have a universal normalized logarithmic surplus $\kappa N^2 + o(N^2)$, where $\kappa = 0.7079245703\ldots$. In particular, after removing the greatest common divisor of the two complete gap products, the unmatched base-3 quotient is larger than $2^{\binom{N}{2}}$ for every $N \geq 2$. This turns the open problem into a precise second-order valuation-transport question.

A separate algebraic calculation gives the exact norm of $E^k - 3^n$ for the canonical radical $E^d = A/3^a$. It proves that using all radical conjugates compresses to a single ordinary binomial gap and cannot, by itself, cross the required scale. These results are substantial new structure, but they do not complete the conjecture.

Contents

1 Executive summary and claim ledger

1.1 Outcome

The conjecture remains open in this report. The work below reaches a new arithmetic object that was not developed in the supplied unified report: the discriminant of the exact compact rotation orbit and the integer products obtained by clearing its denominators. The main gain is not another equivalent formulation of the original four-logarithm determinant. It is an exact decomposition of the "missing cancellation" into synchronized integer gaps and a universal, computable second-order surplus on the base-3 side.

Principal new theorem. Assume a counterexample and let $M = 2^\beta$, $A = 3^\beta$ be its primitive output pair. Form the complete synchronized gap products

$$\mathcal{Q}_{b,B}(N) = \prod_{d=1}^{N-1} |B^d - b^{\lfloor d\beta \rfloor}|^{N-d-C_{d,N}} |B^d - b^{\lceil d\beta \rceil}|^{C_{d,N}},$$

where $C_{d,N}$ is the exact number of wraps of the length-$(N-d)$ rotation segment by $\{d\beta\}$. Then

$$\log_3 \mathcal{Q}_{3,A}(N) - \log_2 \mathcal{Q}_{2,M}(N) > \binom{N}{2} \log_3 2,$$

and, more precisely, the left side divided by N^2 tends to $\kappa = 0.7079245703208558\ldots$. If $G_N = \gcd(\mathcal{Q}_{2,M}(N), \mathcal{Q}_{3,A}(N))$, then

$$\frac{\mathcal{Q}_{3,A}(N)}{G_N} > 2^{\binom{N}{2}}.$$

All three statements are proved in Sections 8 and 9. **[new proof]**

The final inequality says that a hypothetical counterexample forces at least quadratic exponential mass in prime powers that occur in the base-3 gap product but not in the base-2 gap product. In the branch $3 \nmid A$, this mass is supported entirely on primes outside the fixed set of divisors of $3A$. In the branch $3 \mid A$, an exact cubic structural 3-part can be separated, and the unresolved problem is to control the remaining second-order transport.

1.2 Claim ledger

Statement	Status	Main input
Primitive generator $\mathrm{Sol} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and $M = 2^\beta$, $A = 3^\beta$	**[imported]**	Supplied unified report
Finite-type bound $\lVert q\beta\rVert \geq cq^{-\tau}$	**[imported]**	Baker–Wüstholz/Matveev as imported in the supplied report
Exact wrap and lag factorization of the orbit Vandermonde	**[new proof]**	Floor arithmetic and a two-case factorization
Exact structural-prime denominator and integer numerator product	**[new proof]**	Unique factorization at $b = 2$ or 3
Orbit-polynomial discriminant identity	**[new proof]**	Discriminant of a product of integral linear forms
Logarithmic-energy limit and closed form I_b	**[new proof]**	Finite type, discrepancy, uniform integrability, an elementary series integral
Pointwise matched-gap amplification	**[new proof]**	Monotonicity of $b^u/(b^u - 1)$ in b
Universal surplus $\kappa N^2 + o(N^2)$	**[new proof]**	Energy limit plus exact integer product identity
Quadratic unmatched-prime mass	**[new proof]**	Normalized surplus and a gcd decomposition
Exact canonical-radical norm formula	**[new proof]**	Cyclotomic multiplicities and $X^D - Y^D$
Complete proof of Alaoglu–Erdős	**[open target]**	Requires a new cross-base valuation-transport theorem

2 Scope, imported structure, and non-duplication

2.1 What is imported

We use two results from the supplied report and do not reprove them.

1. **[imported]** If the conjecture is false, there is a unique least nonintegral solution $\beta > 0$ and positive integers
$$M = 2^\beta, \qquad A = 3^\beta$$
such that every solution is uniquely $n + k\beta$ with $n, k \in \mathbb{N}_0$. The pair is affinely primitive and
$$\min\{v_2(M), v_3(A)\} = 0. \tag{1}$$
In particular β is irrational and every $k\beta$ has nonzero fractional part.

2. **[imported]** Because $\beta = \log_2 M$ is a ratio of logarithms of fixed algebraic numbers, a two-logarithm lower bound gives constants $c > 0$ and $\tau > 0$ with
$$\lVert q\beta\rVert \geq cq^{-\tau} \qquad (q \geq 1), \tag{2}$$

where $\|y\|$ is distance to the nearest integer. The constants may be very poor; only finiteness of τ is used.

Set throughout

$$\vartheta = \log_2 3, \qquad A = M^{\vartheta}. \tag{3}$$

2.2 What is not repeated

The supplied report already treats the four-exponentials reduction, primitive odd cores, radical degree, exact counting, block equidistribution, arbitrary-node and semigroup Vandermonde determinants, p-adic assignment minima, exact-grid HCIZ estimates, Loewner and Smith-profile certificates, polynomial and Padé approximation, compact S-integral orbits, Sturmian and carry languages, natural boundaries, special functions, and twenty-seven independent continuation reports. Repackaging any of those arguments would violate the non-duplication requirement.

The present object is different. We retain the exact cyclic order of the compact orbit and multiply *all pairwise differences*. This creates an integer discriminant after clearing known denominators, and it allows the base-2 and base-3 products to be compared one gap at a time.

3 Public reconnaissance on the reported AI breakthroughs

This section is not used in any proof. It records what was publicly checkable on August 22, 2026, and separates mathematical verification from attribution.

3.1 Jacobian conjecture

A public exact map in dimension three is accompanied by independent symbolic audits and Lean formalizations. One convenient form is

$$\begin{aligned}
F_1 &= (1+xy)^3 z + y^2(1+xy)(4+3xy),\\
F_2 &= y + 3x(1+xy)^2 z + 3xy^2(4+3xy),\\
F_3 &= 2x - 3x^2 y - x^3 z,
\end{aligned}$$

with $\det JF = -2$ and an explicit three-point fiber. The public provenance records credit Fable for work on the construction; the finite identities can be checked independently. This refutes the standard conjecture in dimensions at least three. The plane case remains open. See the exact audit [8], the Fong–Fable status note [9], and a Lean development [10]. **[public-status audit]**

3.2 Hadamard matrices

The public tracker records a coordinated release covering the previously unresolved admissible orders below 2000, with exact $\{\pm 1\}$ matrices and checker scripts. The tracker pages did not present a completely stable review label at the time of access: one page described partial independent reproduction and review pending, while the later project summary described all listed orders as verified. Accordingly, this report treats the result as a strong public candidate with exact artifacts, not as a substitute for a refereed classification theorem. See [11]. **[public-status audit]**

3.3 Rank at least thirty

The public elliptic-curve leaderboard contains an explicit curve, thirty displayed independent rational points, and an unconditional lower bound rank ≥ 30 for submission #273 [12]. The submitter is listed only as "ranksunbounded." I found no primary public source that identifies that submitter with Fable. The mathematical artifact is therefore public; the model attribution in the prompt is not independently verified here. **[public-status audit]**

3.4 Alaoglu–Erdős

No indexed public proof, preprint, repository, or Lean development of the claimed new Alaoglu–Erdős route was located in the searches performed for this report. That is a search result, not evidence that no private or newly unindexed argument exists. The work below was therefore reconstructed independently.

4 The compact orbit and exact carry calculus

Assume from now on that a counterexample exists. For $k \geq 0$ define

$$r_k = \lfloor k\beta \rfloor, \qquad t_k = \{k\beta\}, \tag{4}$$

and, for a base $b > 1$,

$$X_{b,k} = b^{t_k}. \tag{5}$$

For the two arithmetic bases,

$$X_{2,k} = \frac{M^k}{2^{r_k}}, \qquad X_{3,k} = \frac{A^k}{3^{r_k}}. \tag{6}$$

Thus both coordinates use the same floor sequence and lie in fixed compact intervals.

4.1 Lag data

Fix $1 \leq d < N$. Put

$$h_d = \lfloor d\beta \rfloor, \qquad \eta_d = \{d\beta\}, \qquad L_d = N - d. \tag{7}$$

For $0 \leq i < L_d$, define the carry

$$\varepsilon_{i,d} = r_{i+d} - r_i - h_d. \tag{8}$$

Lemma 4.1 (Exact two-valued carry). *For every admissible i, d,*

$$\varepsilon_{i,d} \in \{0, 1\}, \qquad t_{i+d} = t_i + \eta_d - \varepsilon_{i,d}.$$

Moreover $\varepsilon_{i,d} = 1$ if and only if $t_i \geq 1 - \eta_d$.

Proof. Write $i\beta = r_i + t_i$ and $d\beta = h_d + \eta_d$. Then

$$(i + d)\beta = (r_i + h_d) + (t_i + \eta_d),$$

where $0 < t_i + \eta_d < 2$. Taking the integer and fractional parts gives exactly the stated formulas. □

Define the exact wrap count

$$C_{d,N} = \sum_{i=0}^{L_d-1} \varepsilon_{i,d}. \tag{9}$$

It has the purely integral expression

$$C_{d,N} = \sum_{i=0}^{N-d-1} (r_{i+d} - r_i - h_d). \tag{10}$$

No discrepancy estimate is hidden in this identity.

4.2 Rotation Vandermonde products

Definition 4.2 (Unsquared rotation discriminant)**.** For $b > 1$ and $N \geq 2$, set

$$\Delta_{b,N} = \prod_{0\leq i<j<N} |b^{t_j} - b^{t_i}|. \tag{11}$$

Its square is the usual Vandermonde discriminant of the N orbit points.

Theorem 4.3 (Exact lag factorization)**.** *For every $b > 1$, $1 \leq d < N$, and $L = L_d$,*

$$\prod_{i=0}^{L-1} |b^{t_{i+d}} - b^{t_i}| \tag{12}$$

$$= b^{\sum_{i=0}^{L-1} t_i + C_{d,N}(\eta_d - 1)} (b^{\eta_d} - 1)^{L-C_{d,N}} (b^{1-\eta_d} - 1)^{C_{d,N}}. \tag{13}$$

Consequently $\Delta_{b,N}$ is the product of the right side over $d = 1, \ldots, N-1$.

Proof. If $\varepsilon_{i,d} = 0$, Lemma 4.1 gives

$$|b^{t_{i+d}} - b^{t_i}| = b^{t_i}(b^{\eta_d} - 1).$$

If $\varepsilon_{i,d} = 1$, then $t_{i+d} = t_i + \eta_d - 1 < t_i$ and

$$|b^{t_{i+d}} - b^{t_i}| = b^{t_i+\eta_d-1}(b^{1-\eta_d} - 1).$$

Multiplying the $L - C_{d,N}$ nonwrap factors and the $C_{d,N}$ wrap factors gives (13). Every pair $i < j$ occurs uniquely at lag $d = j - i$, proving the last assertion. □

Remark 4.4. The factors $b^{\eta_d} - 1$ and $b^{1-\eta_d} - 1$ are a two-branch analogue of a Sudler product. The exact exponent $C_{d,N}$ retains the carry word instead of replacing it by its expected value. This is the feature that later allows an integer gap factorization.

5 Exact denominators and integer numerator products

The orbit points are rational when $B = b^\beta$ is an integer. Their pairwise differences have completely explicit denominators at the structural prime whenever $b \nmid B$.

5.1 One pair

Let $b \in \{2,3\}$, let $B = b^\beta \in \mathbb{N}$, and write

$$X_k = \frac{B^k}{b^{r_k}} = b^{t_k}.$$

For $i < j$,

$$X_j - X_i = \frac{B^j - B^i b^{r_j - r_i}}{b^{r_j}} = \frac{B^i(B^{j-i} - b^{r_j - r_i})}{b^{r_j}}. \tag{14}$$

Theorem 5.1 (Exact structural-prime denominator)**.** *Assume $b \nmid B$. Then the numerator in (14) is a b-adic unit, so the reduced denominator of $X_j - X_i$ is exactly b^{r_j}.*

Proof. The imported finite verification implies $\beta > 1$; only $\beta > 1$ is needed here. Thus $r_j - r_i \geq 1$. Modulo b, the term B^{j-i} is nonzero while $b^{r_j - r_i}$ is zero. Multiplication by the b-adic unit B^i preserves nonvanishing. □

5.2 The complete numerator

Define

$$E_N = \sum_{j=1}^{N-1} j r_j \tag{15}$$

and

$$P_{b,B}(N) = \prod_{0 \leq i < j < N} |B^j - B^i b^{r_j - r_i}| \in \mathbb{N}. \tag{16}$$

Proposition 5.2 (Cleared Vandermonde)**.** *For every $b \in \{2,3\}$ and integer $B = b^\beta$,*

$$P_{b,B}(N) = b^{E_N} \Delta_{b,N}. \tag{17}$$

If $b \nmid B$, then $P_{b,B}(N)$ is coprime to b and

$$v_b(\Delta_{b,N}) = -E_N.$$

Moreover

$$E_N = \frac{\beta}{3} N^3 + O_\beta(N^2). \tag{18}$$

Proof. Multiplying (14) over all pairs contributes r_j once for each $i = 0, \ldots, j-1$, hence the denominator exponent is E_N. The coprimality assertion follows from Theorem 5.1. Finally,

$$E_N = \beta \sum_{j=1}^{N-1} j^2 - \sum_{j=1}^{N-1} j t_j,$$

and the second sum is $O(N^2)$ while $\sum_{j<N} j^2 = N^3/3 + O(N^2)$. □

Cubic denominator tax. The known structural denominator of the complete orbit Vandermonde has logarithm $\frac{1}{3}\beta(\log b)N^3 + O(N^2)$, whereas the entire archimedean Vandermonde has logarithm only of order N^2 (Theorem 7.2). Therefore an integer-lower-bound argument using only this denominator misses the analytic signal by one full power of N. This is a different deficit from the one-logarithm node-count deficit in the supplied report. **[new proof]**

5.3 Gap-only factorization

The factors B^i in (16) carry no near-power information. Remove them by defining

$$\mathcal{Q}_{b,B}(N) := \prod_{d=1}^{N-1} |B^d - b^{h_d}|^{L_d - C_{d,N}} |B^d - b^{h_d+1}|^{C_{d,N}}. \tag{19}$$

Because $d\beta$ is never an integer, every factor is a positive integer.

Theorem 5.3 (Exact integer gap product)**.** *For every arithmetic pair (b, B) as above,*

$$P_{b,B}(N) = B^{\binom{N}{3}} \mathcal{Q}_{b,B}(N). \tag{20}$$

Proof. At lag d, equation (16) contains

$$B^i |B^d - b^{h_d+\varepsilon_{i,d}}|$$

for $i = 0, \ldots, L_d - 1$. The wrap count supplies the exponents in (19). The total exponent of B is

$$\sum_{d=1}^{N-1} \sum_{i=0}^{N-d-1} i = \sum_{L=1}^{N-1} \frac{L(L-1)}{2} = \binom{N}{3}.$$

□

6 Orbit polynomials and discriminants

The same data can be packaged into one integer polynomial.

Definition 6.1 (Orbit polynomial)**.** For $N \geq 1$, put

$$F_{b,B,N}(X) = \prod_{k=0}^{N-1} (b^{r_k} X - B^k) \in \mathbb{Z}[X]. \tag{21}$$

Its roots are the compact orbit points $X_k = B^k / b^{r_k} = b^{t_k}$.

Let

$$R_N = \sum_{k=0}^{N-1} r_k, \qquad S_N = \sum_{i=0}^{N-2} (N-1-i) r_i. \tag{22}$$

Proposition 6.2 (Endpoint data and discriminant)**.** *The leading and constant coefficients are*

$$\mathrm{lc}(F_{b,B,N}) = b^{R_N}, \qquad |F_{b,B,N}(0)| = B^{\binom{N}{2}}. \tag{23}$$

The discriminant is the positive square

$$|\operatorname{Disc} F_{b,B,N}| = \prod_{0 \leq i < j < N} |B^j b^{r_i} - B^i b^{r_j}|^2 \tag{24}$$

$$= b^{2S_N} P_{b,B}(N)^2. \tag{25}$$

Proof. The endpoint formulas are immediate from the product. For a product of linear forms $\prod_i(q_iX-p_i)$, direct substitution in the root formula for the discriminant gives

$$\mathrm{Disc}\Big(\prod_i(q_iX-p_i)\Big)=\prod_{i<j}(p_iq_j-p_jq_i)^2.$$

Take $p_i=B^i$ and $q_i=b^{r_i}$. Each cross determinant contains b^{r_i} times the corresponding factor in $P_{b,B}(N)$, and r_i occurs $N-1-i$ times. □

For a hypothetical Alaoglu–Erdős pair, the two orbit polynomials have identical endpoint data after normalizing logarithms by the base:

$$\log_3 \mathrm{lc}(F_{3,A,N})=\log_2 \mathrm{lc}(F_{2,M,N})=R_N, \tag{26}$$

$$\log_3|F_{3,A,N}(0)|=\log_2|F_{2,M,N}(0)|=\beta\binom{N}{2}. \tag{27}$$

Their discriminants, however, have a universal positive second-order separation. That is proved after the logarithmic-energy calculation.

7 Logarithmic energy of the orbit

7.1 A finite-type uniform-integrability lemma

For $z_k=t_k=\{k\beta\}$, the empirical measures $\mu_N=N^{-1}\sum_{k<N}\delta_{z_k}$ converge weakly to Lebesgue measure on $[0,1]$. The logarithmic kernel is singular on the diagonal, so weak convergence alone does not justify passing to pairwise energy. The finite-type bound (2) supplies exactly the missing uniform integrability.

Lemma 7.1 (Near-collision tail)**.** *Assume* (2)*. There is* $\delta>0$ *such that, uniformly for intervals* $I\subset[0,1]$,

$$\#\{1\le d<N:\{d\beta\}\in I\}=N|I|+O(N^{1-\delta}). \tag{28}$$

Consequently

$$\lim_{T\to\infty}\limsup_{N\to\infty}\frac{1}{N^2}\sum_{\substack{0\le i\ne j<N\\ |z_i-z_j|<e^{-T}}}-\log|z_i-z_j|=0. \tag{29}$$

Proof. The discrepancy estimate is the standard Erdős–Turán consequence of finite type; see Kuipers–Niederreiter [5]. We include the tail reduction because it is the point at which the hypothesis is used.

Let $A_N(\rho)=\#\{1\le d<N:\|d\beta\|<\rho\}$. Equation (28) gives $A_N(\rho)\le 2\rho N+O(N^{1-\delta})$. If $|z_i-z_j|<\rho$, then $\|(j-i)\beta\|<\rho$, and for each lag there are at most N ordered pairs. The spacing bound (2) also gives $|z_i-z_j|\ge cN^{-\tau}$. Using $(-\log y-T)_+=\int_T^{-\log y}du$ therefore bounds the normalized tail by

$$\int_T^{\tau\log N+O(1)}\Big(4e^{-u}+O(N^{-\delta})\Big)\,du\le 4e^{-T}+O(N^{-\delta}\log N).$$

First let $N\to\infty$, then $T\to\infty$. □

7.2 Energy limit

Theorem 7.2 (Rotation-discriminant energy). *For every $b > 1$,*

$$\frac{2}{N^2}\log\Delta_{b,N} \longrightarrow I_b, \qquad I_b = \int_0^1\int_0^1 \log|b^s - b^t|\,ds\,dt. \tag{30}$$

The constant has the closed form

$$I_b = \frac{2\log b}{3} - \frac{\pi^2}{3\log b} + \frac{2(\zeta(3) - \mathrm{Li}_3(1/b))}{(\log b)^2}. \tag{31}$$

Proof. Set

$$K_b(s,t) = \log|b^s - b^t|.$$

The difference

$$H_b(s,t) = K_b(s,t) - \log|s-t|$$

extends continuously to the diagonal, with $H_b(s,s) = \log((\log b)b^s)$. Weak convergence of $\mu_N \times \mu_N$ handles H_b. For the logarithmic term, first truncate at height $-T$, use weak convergence for the bounded continuous truncation, and then apply Lemma 7.1. The N omitted diagonal terms contribute $O(N)/N^2$ after truncation. This proves (30).

For the integral, let $L = \log b$ and use the density $2(1-u)$ of $u = |s-t|$. Since

$$\log|b^s - b^t| = L\min(s,t) + Lu + \log(1 - e^{-Lu}),$$

the first two expectations total $2L/3$. Expanding $\log(1-e^{-Lu}) = -\sum_{n\ge1} e^{-nLu}/n$ and integrating termwise gives

$$2\int_0^1 (1-u)\log(1-e^{-Lu})\,du = -\frac{\pi^2}{3L} + \frac{2(\zeta(3) - \mathrm{Li}_3(e^{-L}))}{L^2},$$

which is (31). □

7.3 Numerical values

High-precision evaluation gives

$$I_2 = -1.51660817273459213659\ldots, \tag{32}$$
$$I_3 = -0.84829781726239522907\ldots. \tag{33}$$

These negative values are expected: many orbit points are closer than one in the ordinary metric, so the product of differences is exponentially small at the N^2 scale.

For an implementation check, take $\beta = \log_2 5$. The following table lists $2N^{-2}\log\Delta_{b,N}$ and the corresponding normalized base surplus.

N	base 2 energy	base 3 energy	normalized surplus N^{-2}
50	−1.4053349594	−0.7474049836	0.6735762212
100	−1.4458192176	−0.7818859890	0.6870864586
200	−1.4834158187	−0.8196667457	0.6970119103
400	−1.4983414907	−0.8318264138	0.7022444033
800	−1.5053190365	−0.8377156849	0.7045973159
1600	−1.5113158350	−0.8433138330	0.7063752646
3200	−1.5133107059	−0.8451095375	0.7069969994
limit	−1.5166081727	−0.8482978173	0.7079245703

[computation] The table is illustrative only; Theorem 7.2 is proved independently of the computation.

8 Matched-gap amplification

This section is the central new arithmetic comparison.

8.1 The universal one-gap function

Define for $0 < u < 1$

$$\Phi(u) = \log_3(3^u - 1) - \log_2(2^u - 1). \tag{34}$$

Lemma 8.1 (Strict gap amplification)**.** *The function Φ is strictly decreasing on $(0, 1)$ and*

$$\Phi(u) > \log_3 2 \qquad (0 < u < 1). \tag{35}$$

Moreover $\Phi(u) \to +\infty$ as $u \downarrow 0$ and $\Phi(u) \to \log_3 2$ as $u \uparrow 1$.

Proof. For $b > 1$,

$$\frac{d}{du} \log_b(b^u - 1) = \frac{b^u}{b^u - 1} = \frac{1}{1 - b^{-u}}.$$

For fixed $u > 0$ this is strictly decreasing in b, hence $\Phi'(u) < 0$. The endpoint at one is

$$\log_3(3 - 1) - \log_2(2 - 1) = \log_3 2.$$

The expansion $b^u - 1 = u \log b + O(u^2)$ proves divergence at zero. □

8.2 Every synchronized integer gap

For $d \geq 1$, put $h = \lfloor d\beta \rfloor$ and $\eta = \{d\beta\}$. Then

$$A^d - 3^h = 3^h(3^\eta - 1), \tag{36}$$

$$M^d - 2^h = 2^h(2^\eta - 1). \tag{37}$$

For the ceiling branch, the absolute gaps are $3^h(3 - 3^\eta)$ and $2^h(2 - 2^\eta)$.

Theorem 8.2 (Pointwise matched-gap inequality). *For every $d \ge 1$ and $h \in \{\lfloor d\beta \rfloor, \lceil d\beta \rceil\}$,*

$$|A^d - 3^h| > 2\,|M^d - 2^h|^{\vartheta}. \tag{38}$$

Equivalently,

$$\log_3 |A^d - 3^h| - \log_2 |M^d - 2^h| > \log_3 2. \tag{39}$$

Proof. For $h = \lfloor d\beta \rfloor$, the h terms cancel in the normalized logarithms and the left side of (39) is $\Phi(\eta)$. For $h = \lceil d\beta \rceil$ it is

$$\log_3(3 - 3^{\eta}) - \log_2(2 - 2^{\eta}) = \Phi(1 - \eta),$$

because $b - b^{\eta} = b^{\eta}(b^{1-\eta} - 1)$. Lemma 8.1 proves both cases. Multiplying (39) by $\log 3$ and using $\vartheta = \log_2 3$ gives (38). □

8.3 Complete products

Theorem 8.3 (Finite universal surplus). *For every $N \ge 2$,*

$$\mathcal{G}_N := \log_3 \mathcal{Q}_{3,A}(N) - \log_2 \mathcal{Q}_{2,M}(N) > \binom{N}{2} \log_3 2. \tag{40}$$

Equivalently,

$$\mathcal{Q}_{3,A}(N) > 2^{\binom{N}{2}}\, \mathcal{Q}_{2,M}(N)^{\vartheta}. \tag{41}$$

The same surplus can be written in any of the following exact forms:

$$\begin{aligned}
\mathcal{G}_N &= \log_3 P_{3,A}(N) - \log_2 P_{2,M}(N), && (42)\\
&= \log_3 \Delta_{3,N} - \log_2 \Delta_{2,N}, && (43)\\
&= \sum_{0 \le i < j < N} \Phi(|t_j - t_i|), && (44)\\
&= \sum_{d=1}^{N-1} \big((L_d - C_{d,N})\Phi(\eta_d) + C_{d,N}\Phi(1 - \eta_d)\big). && (45)
\end{aligned}$$

Proof. Apply Theorem 8.2 to each of the $(L_d - C_{d,N})$ floor gaps and $C_{d,N}$ ceiling gaps at lag d. The total number of factors is $\sum_{d=1}^{N-1} L_d = \binom{N}{2}$, giving (40) and (41).

Equation (42) follows from Theorem 5.3: the monomial terms cancel because

$$\log_3 A = \log_2 M = \beta.$$

Equation (43) follows from Proposition 5.2, since the common E_N cancels. For a pair with $s = \min(t_i, t_j)$ and $u = |t_i - t_j|$,

$$\log_b |b^{t_j} - b^{t_i}| = s + \log_b(b^u - 1),$$

so the s terms cancel between bases and leave $\Phi(u)$. This proves (44); grouping by lags gives (45). □

8.4 The exact second-order constant

Theorem 8.4 (Universal matched-gap energy)**.** *One has*

$$\frac{\mathcal{G}_N}{N^2} \longrightarrow \kappa, \qquad \kappa = \frac{1}{2}\left(\frac{I_3}{\log 3} - \frac{I_2}{\log 2}\right), \tag{46}$$

where

$$\kappa = \frac{\pi^2}{6}\left(\frac{1}{(\log 2)^2} - \frac{1}{(\log 3)^2}\right) \tag{47}$$

$$+ \frac{\zeta(3) - \mathrm{Li}_3(1/3)}{(\log 3)^3} - \frac{\zeta(3) - \mathrm{Li}_3(1/2)}{(\log 2)^3} \tag{48}$$

$$= 0.70792457032085580787744\ldots. \tag{49}$$

In particular

$$\kappa > \frac{1}{2}\log_3 2 = 0.3154648767\ldots. \tag{50}$$

Proof. Combine (43) with Theorem 7.2. Substitution of (31) gives the closed form; the 2/3 terms cancel. Inequality (50) also follows without numerical evaluation by dividing (40) by N^2 and taking the limit. □

Corollary 8.5 (Discriminant separation)**.** *The orbit polynomials satisfy*

$$\log_3 |\operatorname{Disc} F_{3,A,N}| - \log_2 |\operatorname{Disc} F_{2,M,N}| = 2\mathcal{G}_N > 2\binom{N}{2}\log_3 2, \tag{51}$$

while their normalized leading and constant coefficients agree exactly by (26)–(27). The left side divided by N^2 tends to 2κ.

Proof. Use Proposition 6.2. The common S_N cancels, leaving twice (42). □

9 Quadratic unmatched-prime mass

The universal surplus has an immediate arithmetic interpretation that does not mention real powers.

Theorem 9.1 (Unmatched base-3 quotient)**.** *Let*

$$H_N = \gcd(\mathcal{Q}_{2,M}(N), \mathcal{Q}_{3,A}(N)), \qquad Y_N = \frac{\mathcal{Q}_{3,A}(N)}{H_N}. \tag{52}$$

Then for every $N \geq 2$,

$$Y_N > 2^{\binom{N}{2}}. \tag{53}$$

More precisely,

$$\liminf_{N\to\infty} \frac{\log Y_N}{N^2} \geq \kappa \log 3 = 0.7777346324045835\ldots. \tag{54}$$

Proof. Write $\mathcal{Q}_{2,M} = H_N X_N$ and $\mathcal{Q}_{3,A} = H_N Y_N$ with $\gcd(X_N, Y_N) = 1$. Since $\log 3 > \log 2$,

$$\begin{aligned}\mathcal{G}_N &= \log H_N \left(\frac{1}{\log 3} - \frac{1}{\log 2}\right) + \log_3 Y_N - \log_2 X_N \\ &\le \log_3 Y_N.\end{aligned}$$

Apply Theorems 8.3 and 8.4. □

This is stronger than saying that the two gap products are unequal. It forces a rapidly growing quotient supported on prime powers absent from the dyadic gap product.

9.1 Support in the triadically primitive branch

Write

$$A = 3^b V, \qquad b = v_3(A)\,, \qquad 3 \nmid V. \tag{55}$$

Corollary 9.2 (External support when $3 \nmid A$)**.** *If $b = 0$, then*

$$\gcd(\mathcal{Q}_{3,A}(N), 3A) = 1. \tag{56}$$

Consequently the quotient Y_N in Theorem 9.1 is supported on primes outside the fixed set of divisors of $3A$ and has logarithmic size at least $(\kappa \log 3 + o(1))N^2$.

Proof. Every factor of $\mathcal{Q}_{3,A}$ is $|A^d - 3^h|$ with $h \ge 1$. Modulo 3 it is nonzero because $3 \nmid A$. If $p \mid A$ and $p \ne 3$, then $A^d - 3^h \equiv -3^h \not\equiv 0 \pmod p$. □

9.2 Exact structural part when $3 \mid A$

If $b > 0$, the 3-part of the triadic gap product is cubic and must be separated before the quadratic surplus can be interpreted as genuinely new primes. Put

$$\gamma = \beta - b = \log_3 V > 0.$$

Because γ is irrational, $\lceil d\gamma \rceil \ge 1$ for every $d \ge 1$.

Proposition 9.3 (Exact structural 3-part)**.** *If $b > 0$, then*

$$v_3(\mathcal{Q}_{3,A}(N)) = b\binom{N+1}{3} + \sum_{\substack{1 \le d < N \\ d\gamma < 1}} (N - d - C_{d,N})\, v_3\!\left(V^d - 1\right). \tag{57}$$

The sum on the right has finitely many d, independent of N, and is $O_A(N)$. Hence

$$v_3(\mathcal{Q}_{3,A}(N)) = \frac{b}{6} N^3 + O_A(N^2). \tag{58}$$

Proof. For $h = \lceil d\beta \rceil = bd + \lceil d\gamma \rceil > bd$,

$$v_3\!\left(A^d - 3^h\right) = bd.$$

For $h = \lfloor d\beta \rfloor = bd + \lfloor d\gamma \rfloor$ the same holds unless $\lfloor d\gamma \rfloor = 0$. In that exceptional case the valuation is $bd + v_3\Big(V^d - 1\Big)$. The floor branch appears with exponent $N - d - C_{d,N}$. Finally

$$\sum_{d=1}^{N-1} d(N-d) = \binom{N+1}{3}.$$

□

There is a symmetric statement for $a = v_2(M) > 0$. The primitive-depth condition (1) says that the two structural cubic terms never occur simultaneously. The open problem is not to discover these terms; they are exact. It is to transport the remaining prime mass across the two bases at the N^2 scale.

9.3 The valuation-transport target

Expanding the universal surplus into prime valuations gives the exact identity

$$\mathcal{G}_N = \sum_p \left(\frac{v_p(\mathcal{Q}_{3,A}(N))}{\log 3} - \frac{v_p(\mathcal{Q}_{2,M}(N))}{\log 2} \right) \log p. \tag{59}$$

This is merely unique factorization, but it pinpoints what a closing theorem must do.

Criterion 9.4 (Cross-base valuation transport)**.** *Any arithmetic theorem, applicable to every primitive integer pair* (M, A)*, which forces*

$$\limsup_{N\to\infty} \frac{1}{N^2} \sum_p \left(\frac{v_p(\mathcal{Q}_{3,A}(N))}{\log 3} - \frac{v_p(\mathcal{Q}_{2,M}(N))}{\log 2} \right) \log p < \kappa \tag{60}$$

would prove the Alaoglu–Erdős conjecture.

Proof. Under a counterexample, Theorem 8.4 says that the same expression tends to κ. □

Criterion 9.4 is intentionally phrased as a sufficient condition, not as a new name for the conjecture. Its value is that every summand concerns valuations of explicit ordinary integer differences

$$M^d - 2^{\lfloor d\beta \rfloor}, \quad M^d - 2^{\lceil d\beta \rceil}, \quad A^d - 3^{\lfloor d\beta \rfloor}, \quad A^d - 3^{\lceil d\beta \rceil}.$$

There are no logarithmic coefficients inside those valuations. The transcendental relation enters only through the common Beatty exponents.

Audit warning 9.5. A small-gcd theorem by itself does not close the problem. Standard subspace-theorem estimates for common divisors tend to make H_N smaller, which makes the unmatched quotient in Theorem 9.1 larger. The needed input is a cross-base *upper* control on the normalized unshared mass, or a mechanism that converts its growth into a forbidden third algebraic exponential.

10 Canonical-radical norm compression

The supplied report reduces a counterexample to a canonical radical. We now compute exactly what all of its conjugates contribute to a near-power gap.

10.1 Setup

[imported] There are integers $w > 1$ odd, $d \geq 1$, $a \geq 0$, and $A \geq 1$ such that

$$E = w^{\vartheta} > 0, \qquad E^d = q = \frac{A}{3^a}, \tag{61}$$

d is the least positive exponent for which $E^d \in \mathbb{Q}$, and $X^d - q$ is irreducible over $\mathbb{Q}$.

Fix $k \geq 1$ and $n \geq 0$. Put

$$g = \gcd(k, d), \qquad K = \frac{k}{g}, \qquad D = \frac{d}{g}. \tag{62}$$

Theorem 10.1 (Exact cyclotomic norm)**.** *The field norm is*

$$\left| N_{\mathbb{Q}(E)/\mathbb{Q}}(E^k - 3^n) \right| = \left| q^K - 3^{nD} \right|^g \tag{63}$$

$$= \frac{|A^K - 3^{aK+nD}|^g}{3^{ak}}. \tag{64}$$

The numerator is nonzero. Furthermore

$$|E^k - 3^n| = \frac{|A^K - 3^{aK+nD}|}{3^{aK} \sum_{j=0}^{D-1} (E^k)^{D-1-j} (3^n)^j}, \tag{65}$$

and hence

$$|E^k - 3^n| \geq \frac{3^{-aK}}{D \max(E^k, 3^n)^{D-1}}. \tag{66}$$

Proof. The conjugates of E are $\zeta_d^j E$. The multiset of kth powers ζ_d^{jk} contains every Dth root of unity exactly g times. Therefore

$$\prod_{j=0}^{d-1} ((\zeta_d^j E)^k - 3^n) = \pm((E^k)^D - 3^{nD})^g = \pm(q^K - 3^{nD})^g.$$

Substitution of $q = A/3^a$ gives (64). If its numerator vanished, positivity would give $E^k = 3^n$, and then $k\vartheta \log w = n \log 3$, so $w^k = 2^n$, impossible for odd $w > 1$.

Finally use

$$X^D - Y^D = (X - Y) \sum_{j=0}^{D-1} X^{D-1-j} Y^j$$

with $X = E^k$, $Y = 3^n$, and clear 3^{aK}. The denominator sum is at most $D \max(X, Y)^{D-1}$. □

10.2 Baker-strengthened relative separation

Because E and 3 are multiplicatively independent positive algebraic numbers, Baker's theorem gives constants $C, \mu > 0$ such that

$$|k \log E - n \log 3| \geq C(k + n)^{-\mu} \tag{67}$$

whenever the form is nonzero. Standard comparison with $e^z - 1$ yields

$$|E^k - 3^n| \geq C' \max(E^k, 3^n)(k+n)^{-\mu'} \tag{68}$$

for suitable $C', \mu' > 0$, after harmless adjustment outside the close regime.

This is stronger than (66), but it still matches the best rational-return scale rather than contradicting it. The norm has not created a new algebraic quantity; it has compressed all conjugates to the ordinary binomial difference $A^K - 3^{aK+nD}$.

Exact no-go statement. Any argument that uses the conjugates of E only through the field norm of $E^k - 3^n$ reduces exactly to one two-logarithm gap. Norm amplification alone cannot create the missing third independent exponential or a super-polynomial separation. A successful radical argument must retain more than the total product of conjugates—for example, a signed subset product, a resolvent sensitive to the real embedding, or a coupled finite-place invariant. **[new proof]**

11 Why the new identities still stop short

11.1 The integer lower bound has the wrong scale

When $b \nmid B$, the rational number $\Delta_{b,N}$ has exact denominator b^{E_N}, so nonvanishing gives only

$$|\Delta_{b,N}| \geq b^{-E_N} = \exp\left(-\frac{\beta \log b}{3} N^3 + O(N^2)\right).$$

The true size is

$$|\Delta_{b,N}| = \exp\left(\frac{I_b}{2} N^2 + o(N^2)\right).$$

The lower bound is therefore exponentially weaker by one power of N. Taking squares, discriminants, or products of the same denominators cannot change that order.

11.2 The universal surplus is compatible with ordinary arithmetic

The inequality

$$\mathcal{Q}_{3,A} > 2^{\binom{N}{2}} \mathcal{Q}_{2,M}^{\vartheta}$$

is strong but points in a direction that large-base geometry naturally permits. Both gap products themselves have logarithm of order N^3; the surplus is only their second-order difference. An arbitrary pair of unrelated integer products can easily have such a surplus. The conjecture can be reached only by exploiting the fact that both products use the *same* Beatty exponents and arise from the same primitive output pair.

11.3 Why a gcd upper bound is not enough

Theorem 9.1 already says that the quotient remaining after the gcd is enormous. A theorem of Bugeaud–Corvaja–Zannier type [6] that makes the common divisor even smaller does not contradict anything. What is needed is one of the following stronger mechanisms:

1. an upper bound for the weighted unmatched base-3 prime mass at the exact N^2 scale;
2. a rule transporting a fixed positive proportion of that mass into the dyadic gap product;
3. a primitive-divisor theorem whose new primes manufacture a third algebraic exponential;
4. a cross-resultant or resolvent whose product formula turns the surplus into an upper, not a lower, archimedean bound.

11.4 Why the radical norm is not the missing transport

Equation (64) has only one numerator, and Baker controls it polynomially in the exponent. It does not connect primes dividing $M^d - 2^h$ to primes dividing $A^d - 3^h$. Thus the radical and discriminant routes currently meet only at the same unresolved transport problem.

12 Concrete research program

12.1 Priority A: cross-base local orders

For a prime $p \nmid 6MA$, simultaneous divisibility

$$p \mid M^d - 2^h, \qquad p \mid A^d - 3^h$$

means that the reduction of $(M, A)^d(2, 3)^{-h}$ is the identity in $(\mathbb{F}_p^\times)^2$. Fixing a generator of $\mathbb{F}_p^\times$ turns this into a 2×2 modular linear system. The determinant of its discrete-log matrix is the finite place analogue of the real logarithmic determinant. A useful theorem would compare the orders of this kernel for many p with the exact Beatty set $h \in \{\lfloor d\beta \rfloor, \lceil d\beta \rceil\}$.

The key requirement is quantitative: an $o(N^2)$ estimate is not enough if it concerns only the number of primes. It must control the log-weighted valuations in (59).

12.2 Priority B: primitive divisors of moving-exponent gaps

Classical Zsigmondy theory handles $a^n - b^n$. Here the relevant factors are

$$A^d - 3^{\lfloor d\beta \rfloor}, \qquad A^d - 3^{\lceil d\beta \rceil},$$

with two exponents moving along an irrational line. A theorem guaranteeing a primitive prime for most d, together with a criterion excluding that prime from the corresponding dyadic gap, would turn Theorem 9.1 from aggregate mass into a supply of genuinely new prime directions. The second step is essential; primitive divisors on the triadic side alone do not force another algebraic exponential.

12.3 Priority C: signed or partial radical norms

The full norm destroys the real embedding. A more promising radical object is a partial product over a Galois-stable but not full set of conjugates, or a resolvent that records which conjugate lies on the positive real branch. Its denominator and local valuations must still be controlled exactly. Equation (64) is a test: any proposed invariant that collapses to that formula has not added information.

12.4 Priority D: orbit-polynomial comparison

The polynomials $F_{2,M,N}$ and $F_{3,A,N}$ have equal normalized leading and constant coefficients but discriminants separated by $2\kappa N^2 + o(N^2)$. A cross-base coefficient inequality sensitive to complete splitting over $\mathbb{Q}$, rather than to Mahler measure alone, could convert this into a contradiction. Standard discriminant–Mahler bounds are too coarse: their leading scale is N^3 and they tolerate the observed N^2 difference.

12.5 Priority E: formalization before further optimization

The finite identities in Sections 4, 5, 6, 8, and 10 are suitable for immediate Lean formalization. Doing so would protect the exponent bookkeeping—especially the wrap multiplicities and discriminant powers—before a deeper valuation experiment is built on top of them.

13 Lean-oriented formalization blueprint

No claim of kernel verification is made for this continuation. The following decomposition keeps all analytic inputs explicit.

13.1 Module `RotationCarry`

Definitions:

$$r(k) = \lfloor k\beta \rfloor, \quad t(k) = k\beta - r(k), \quad \varepsilon(i,d) = r(i+d) - r(i) - r(d).$$

Targets:

- $\varepsilon(i,d) = 0 \vee \varepsilon(i,d) = 1$;
- the exact fractional-part update;
- the floor-sum formula for $C_{d,N}$.

All are elementary ordered-ring/floor lemmas.

13.2 Module `RotationVandermonde`

Define the finite product $\Delta_{b,N}$ and prove Theorem 4.3 by partitioning the finite index set according to the carry. No transcendence theorem is needed; irrationality of β is used only to make every factor nonzero.

13.3 Module `OrbitGapProduct`

For prime b and natural B, define the integer numerator matrix

$$N_{ij} = B^j - B^i b^{r_j - r_i}.$$

Formalize:

- the exact denominator under $b \nmid B$;
- $P = b^{E_N} \Delta$;
- $P = B^{\binom{N}{3}} \mathcal{Q}$;
- the finite sums $\sum_{j<N} j r_j$ and $\sum_{d<N} \sum_{i<N-d} i = \binom{N}{3}$.

13.4 Module `OrbitPolynomialDiscriminant`

Formalize the general identity

$$\operatorname{Disc}\left(\prod_i (q_i X - p_i)\right) = \prod_{i<j} (p_i q_j - p_j q_i)^2$$

for a field, then specialize to integral coefficients. This module should be independent of real powers once the integer sequences are supplied.

13.5 Module `MatchedGapAmplification`

The key analytic lemma is one-dimensional:

$$u \mapsto \log_3(3^u - 1) - \log_2(2^u - 1)$$

is strictly decreasing. Mathlib's derivative rules for real powers and logarithms should make the proof direct. The integer product inequality and unmatched-gcd corollary are then finite algebra and order arithmetic.

A schematic endpoint theorem is:

```
theorem unmatched_triadic_gap_mass
    (hpair : A = Real.rpow M (Real.log 3 / Real.log 2))
    (hbeta : ... exact floor synchronization ... ) :
    2 ^ Nat.choose N 2 < Q3 N / Nat.gcd (Q2 N) (Q3 N)
```

The actual statement should avoid natural-number division by expressing the quotient through a witness from the gcd divisibility theorem.

13.6 Module `RotationLogEnergy`

Separate the theorem into two interfaces:

1. a finite-type discrepancy hypothesis implying uniform integrability of the logarithmic kernel;
2. the elementary evaluation of I_b.

The finite algebraic conclusions do not depend on this module; only the limit constant κ does.

13.7 Module `CanonicalRadicalNorm`

Formalize the multiset map $j \mapsto jk \pmod d$, its fibers of size $g = \gcd(k, d)$, and the cyclotomic product. The exact norm theorem can be stated first for an abstract root of $X^d - q$ and then specialized to the positive real radical.

14 Computational reconnaissance

The calculations in this report are symbolic. Computation was used only to check indexing and to evaluate constants. The following self-contained Python script verifies the finite product factorization for exact integers and reproduces the energy table.

Listing 1: Exact product checks and numerical energy.

```python
from __future__ import annotations

from math import comb, floor, log
from typing import List

import mpmath as mp
import numpy as np

mp.mp.dps = 100


def floor_sequence(base: int, output: int, n: int) -> List[int]:
    """r_k=floor(k log_base(output)), evaluated at high precision."""
    beta = mp.log(output) / mp.log(base)
    return [int(mp.floor(k * beta)) for k in range(n)]


def direct_numerator_product(base: int, output: int, n: int) -> int:
    r = floor_sequence(base, output, n)
    ans = 1
    for i in range(n):
        for j in range(i + 1, n):
            ans *= abs(output**j - output**i * base ** (r[j] - r[i]))
    return ans


def grouped_gap_product(base: int, output: int, n: int) -> int:
    r = floor_sequence(base, output, n)
    q = 1
    for d in range(1, n):
        h = r[d]
        wraps = sum(r[i + d] - r[i] - h for i in range(n - d))
        q *= abs(output**d - base**h) ** (n - d - wraps)
        q *= abs(output**d - base ** (h + 1)) ** wraps
```

```python
    return output ** comb(n, 3) * q


def verify_exact(base: int, output: int, n: int) -> None:
    assert output % base != 0
    left = direct_numerator_product(base, output, n)
    right = grouped_gap_product(base, output, n)
    assert left == right

    r = floor_sequence(base, output, n)
    for i in range(n):
        for j in range(i + 1, n):
            numerator = output**j - output**i * base ** (r[j] - r[i])
            assert numerator % base != 0  # exact reduced denominator


def I(base: int) -> mp.mpf:
    L = mp.log(base)
    return (2 * L / 3 - mp.pi**2 / (3 * L)
            + 2 * (mp.zeta(3) - mp.polylog(3, mp.mpf(1) / base)) / L**2)


def empirical_energy(beta: float, base: int, n: int) -> float:
    t = (np.arange(n, dtype=np.float64) * beta) % 1.0
    x = np.power(float(base), t)
    total = 0.0
    for i in range(n):
        total += np.log(np.abs(x[i + 1:] - x[i])).sum()
    return 2.0 * total / (n * n)


if __name__ == "__main__":
    verify_exact(2, 5, 14)
    verify_exact(3, 10, 14)

    beta = log(5, 2)
    for n in [50, 100, 200, 400, 800, 1600, 3200]:
        e2 = empirical_energy(beta, 2, n)
        e3 = empirical_energy(beta, 3, n)
        surplus = 0.5 * (e3 / log(3) - e2 / log(2))
        print(n, e2, e3, surplus)

    kappa = mp.mpf("0.5") * (I(3) / mp.log(3) - I(2) / mp.log(2))
    print("I2 =", I(2))
    print("I3 =", I(3))
    print("kappa =", kappa)
```

The exact check uses $(b, B) = (2, 5)$ and $(3, 10)$ separately; no counterexample is assumed or simulated. The energy test can use any finite-type irrational rotation because the limiting constant depends on b, not on the rotation angle.

15 Conclusions

The attached report identified two recurring obstacles: the 2×2 four-exponentials wall and a shortage of one logarithmic factor in determinant arguments. The present continuation does not remove either wall, but it changes the arithmetic interface in three concrete ways.

First, every compact-orbit Vandermonde is now factored exactly by lag and carry, and its structural denominator is known with equality. This proves that denominator clearing by itself has a cubic tax against a quadratic analytic signal.

Second, comparing the two bases before discarding the carry structure produces a universal pointwise inequality. Its complete product is not merely nonzero: it has a fixed quadratic surplus and forces an exponentially large quotient of triadic gap prime powers not shared by the dyadic gaps. The constant κ is explicit and independent of the hypothetical counterexample.

Third, the full conjugate norm of the canonical radical is computed exactly and shown to collapse to one ordinary binomial gap. This eliminates a broad family of norm-only arguments and specifies what an improved radical invariant must retain.

The clearest next theorem is therefore not another reformulation of $\log M \log 3 = \log A \log 2$. It is a second-order valuation-transport estimate for the explicit moving-exponent gaps in (59). Proving such an estimate with a constant strictly below κ, or converting the forced unmatched primes into a third algebraic exponential, would complete the conjecture.

A Details of the energy integral

For completeness, let $L = \log b$. Symmetry and the change of variables $u = |s - t|$ give

$$\begin{aligned} I_b &= 2\int_0^1 \int_0^{1-u} \Big(Ls + \log(e^{Lu} - 1)\Big)\, ds\, du \\ &= \frac{L}{3} + 2\int_0^1 (1-u)\log(e^{Lu} - 1)\, du \\ &= \frac{2L}{3} + 2\int_0^1 (1-u)\log(1 - e^{-Lu})\, du. \end{aligned}$$

Now

$$\int_0^1 (1-u)e^{-nLu}\, du = \frac{1}{nL} - \frac{1 - e^{-nL}}{n^2L^2}.$$

Therefore

$$\begin{aligned} 2\int_0^1 (1-u)\log(1 - e^{-Lu})\, du &= -2\sum_{n\geq 1} \frac{1}{n}\left(\frac{1}{nL} - \frac{1 - e^{-nL}}{n^2L^2}\right) \\ &= -\frac{2\zeta(2)}{L} + \frac{2(\zeta(3) - \mathrm{Li}_3(e^{-L}))}{L^2}, \end{aligned}$$

which is (31).

B A compact dependency graph

Result	Dependencies
Exact carry and lag factorization	floor addition; no imported transcendence
Exact denominator	b prime, $b \nmid B$, $\beta > 1$
Integer gap product	exact carry and finite product reindexing
Orbit discriminant	general discriminant of integral linear factors
Energy limit	irrational rotation, finite-type bound, discrepancy
Closed form I_b	elementary integration and power series
Pointwise amplification	$A = M^{\vartheta}$ and monotonicity of Φ
Finite surplus	pointwise amplification and exact wrap multiplicities
Limit κ	finite surplus identity plus energy theorem
Unmatched quotient	finite surplus plus gcd factorization
Radical norm	irreducible binomial and cyclotomic multiplicity

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